

Koopman–JEPA

How does a temporal JEPA encode dynamics?

Final report of the controlled experiment

September 2026

When only temporal pairing changes, the learned predictor changes with it. Static, cyclic, and independent laws are identified in 10/10 seeds per condition, and the active spectrum tracks continuous dynamic persistence with spectral-modulus MAE 0.027.

1 Question

We seek a representation $z_t = f_\theta(x_t)$ and a linear predictor M such that

$$Mz_t \approx z_{t+1}.$$

If $\psi(r_t)$ describes the true phase, K is its operator, and the encoder learns $z \approx A\psi$, the relevant test is

$$MA \approx AK.$$

The encoder may choose any latent basis, so we do not require $M = K$. We compare both the action of M and its spectrum restricted to the active latent span.

Theoretical motivation. Ruiz-Morales et al. show that, under idealized assumptions, a time-series JEPA can encode dynamical-regime indicators: Koopman eigenfunctions with $\lambda = 1$. A near-identity linear predictor selects this interpretable invariant solution from equivalent optima. This experiment asks the natural next question: can the same predictive architecture move beyond invariants and recover nontrivial modes such as $\{-1, i, -i\}$ and $\rho\{-1, i, -i\}$? See Ruiz-Morales et al., “Koopman Invariants as Drivers of Emergent Time-Series Clustering in Joint-Embedding Predictive Architectures,” AAAI 2026.

2 Controlled observations

The hidden state has four phases $r \in \{0, 1, 2, 3\}$. Each phase emits a length-128 window:

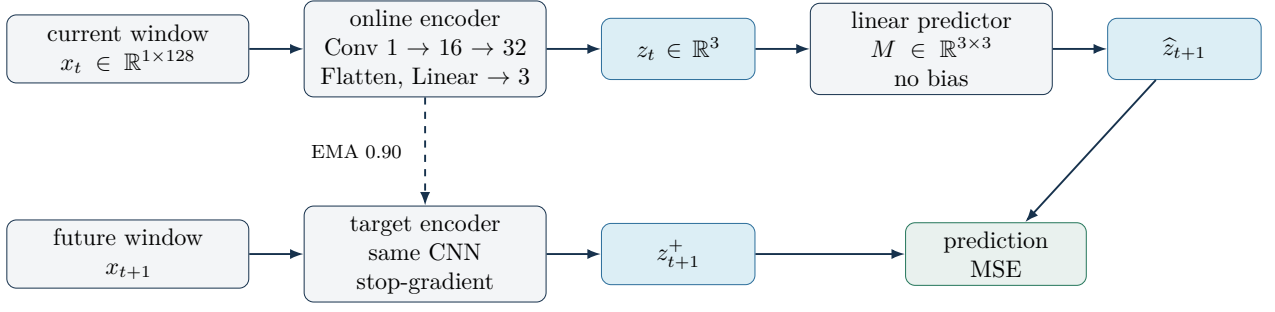
$$x_t[\tau] = a_t s_{r_t}(\tau - \delta_t) + b_t + \epsilon_{t,\tau},$$

with random amplitude, offset, temporal jitter, and noise. Every condition reuses exactly the same current and future window banks; only temporal pairing changes.

Dynamics	Transition	Active spectrum
Static	$r_{t+1} = r_t$	$\{1, 1, 1\}$
Cyclic	$r_{t+1} = r_t + 1 \pmod{4}$	$\{-1, i, -i\}$
Independent	uniform future phase	$\{0, 0, 0\}$

Each seed and condition contains 1,024 training pairs and 256 validation pairs. The main comparison uses 10 paired seeds. The observation audit verifies identical single-window marginals, correct transitions, and perfectly decodable phase.

3 Architecture and assumptions



Training runs for 60 epochs with AdamW. The predictor learning rate is $4\times$ the encoder rate; the online encoder is frozen after epoch 3 and the target follows an EMA update. Mean, variance, and covariance penalties prevent collapse.

The 3D latent matches the true dynamic rank, and the early encoder freeze stabilizes predictor optimization. Both are part of the experimental scope.

4 The predictor follows the transition law

For each candidate d , we compute

$$E_d = \frac{\|MA - AK_d\|_F}{\|A\|_F}.$$

We also compare the eigenvalues of M on the active span. The predeclared criterion is at least 8/10 correct seeds by action and spectrum. Training loss does not enter this decision.

Dynamics	Action	Spectrum	Error	Margin
Static	10/10	10/10	0.142	0.806
Cyclic	10/10	10/10	0.068	0.877
Independent	10/10	10/10	0.015	0.981

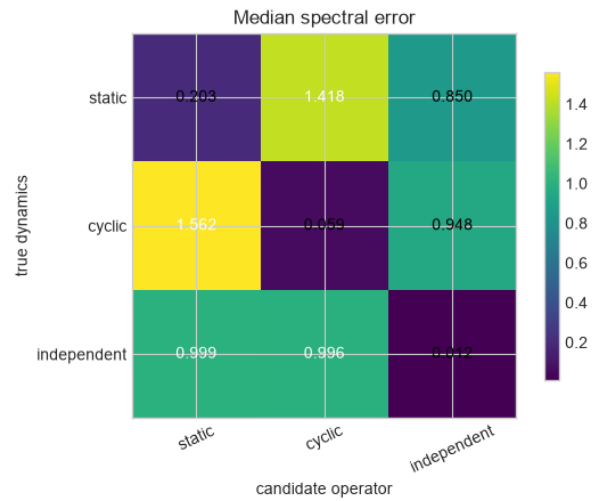
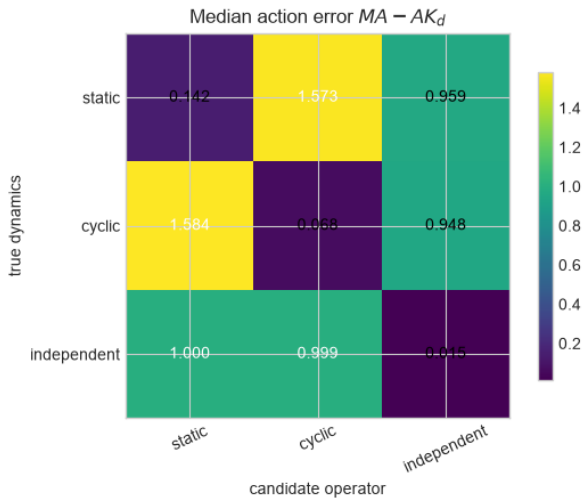


Figure 1: Median error against each candidate. Rows differ only in temporal pairing. The low diagonal shows that the predictor follows the transition law rather than single-window statistics.

5 One-step and multi-step behavior

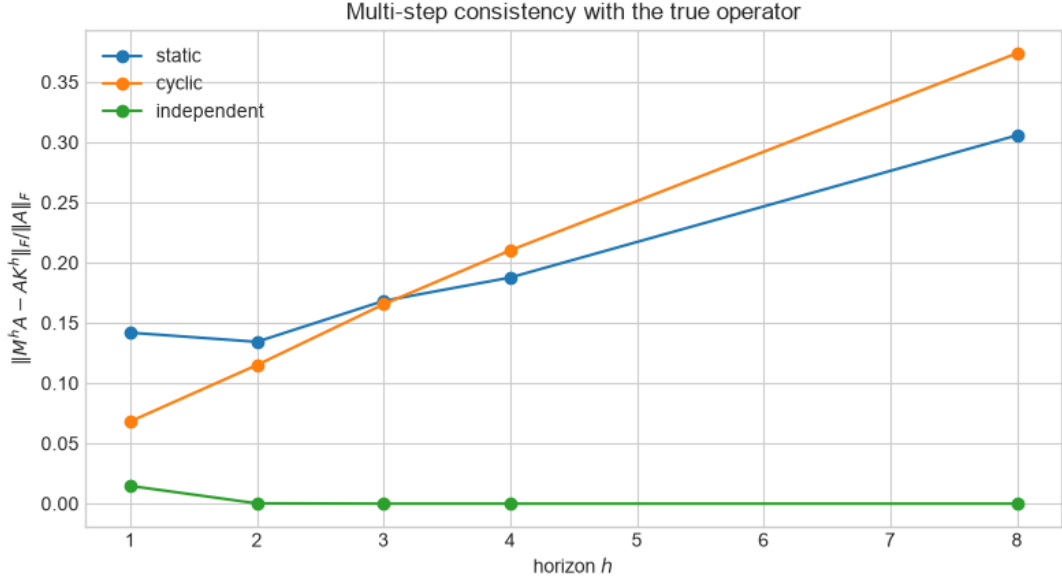


Figure 2: Error $\|M^h A - AK^h\|_F / \|A\|_F$. Independent dynamics contract rapidly; static and cyclic errors accumulate with horizon.

The learned cyclic action contains the nontrivial active spectrum $\{-1, i, -i\}$. Identification is strongest at one step. Repeated application of M magnifies small discrepancies, so the experiment does not claim perfect long-horizon prediction.

6 Continuous dynamic persistence

We interpolate between the cyclic law C and a uniform law U :

$$P_\rho = \rho C + (1 - \rho)U, \quad K_\rho = \rho C, \quad \sigma(K_\rho) = \rho\{-1, i, -i\}.$$

ρ	Action	Spectrum	Median modulus
0.00	10/10	10/10	0.016
0.25	10/10	10/10	0.239
0.50	9/10	9/10	0.476
0.75	8/10	8/10	0.718
1.00	7/10	7/10	0.948

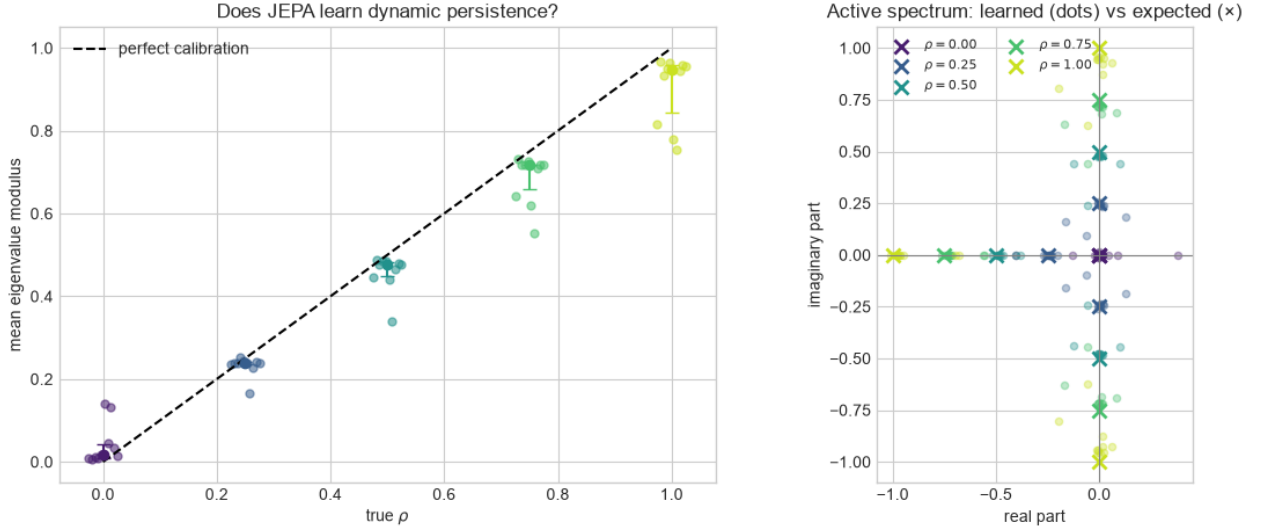


Figure 3: Left: learned spectral modulus against persistence. Right: expected crosses and learned eigenvalues in the complex plane.

Continuous calibration has MAE 0.027, slope 0.937, intercept 0.011, and $R^2 = 0.9998$. The predictor is slightly contractive at high persistence. Some seeds therefore choose the neighboring grid value, and the strict 8/10 criterion does not pass at $\rho = 1$. This is an observed behavior, not replaced by a post-hoc threshold.

7 Conclusion

What the experiment establishes:

- JEPA does not collapse to an invariant representation.
- The predictor changes when only temporal pairing changes.
- Its active action and spectrum recover three qualitatively different laws.
- Its spectral radius tracks continuous persistence.
- Small one-step errors accumulate in multi-step rollouts.

Scope: the result depends on this observation process, 3D latent, early encoder freeze, EMA, and anti-collapse regularization. It does not imply universal Koopman recovery or robustness to different observations and state spaces. It is a controlled characterization of this JEPA’s learned dynamics.

Reproducibility

The argument is contained in four executed notebooks under `notebooks/`. Exact configurations are in `configs/`, the consolidated design is in `docs/EXPERIMENT.md`, and reusable implementation code is in `src/koopman_jepa/`. Tests cover data generation, operator algebra, training, and committed notebook outputs.